Sparse Dictionary Learning for Interpretable Lung Disease Diagnosis from 3D Chest CT

An undergraduate project proposal report submitted to the

Department of Electrical and Information Engineering
Faculty of Engineering
University of Ruhuna
Sri Lanka

in partial fulfillment of the requirements for the

Degree of the Bachelor of the Science of Engineering Honours

by

K.R.C.S. Kekulawala	-	EG/2021/4609
M.D.Y.B. Kularathne	-	EG/2021/4620
B.G.P.U. Thilakshana	-	EG/2021/4831
B.P.G.L. Thejana	-	EG/2021/4828

....

Dr. Kaveen Liyanage (Supervisor)

Mrs. Sithara Mahagama (Co-Supervisor)

Abstract

Abstract goes here.

Contents

List of Figures

List of Tables

Acronyms

Chapter 1

Introduction

Chest Computed Tomography (CT) currently serves as the definitive modality for assessing a wide range of lung pathologies. Modern radiology increasingly relies on artificial intelligence to streamline the acquisition of radiologist analyses on chest X-rays, as evidenced by a study wherein an AI system reduced interpretation delivery times from 11.2 days to a mere 2.7 days, reinforcing the potency of automated triaging systems in streamlining healthcare workflows and amplifying patient care standards [42].[1]

1.1 Background

Lung disease remains a leading cause of global morbidity and mortality. In the diagnosis of lung disease, assessing the pattern, location, and regional distribution of involvement is the domain of radiology. Clinical diagnosis increasingly relies on radiological patterns to differentiate between pathologies with overlapping symptoms. While Chest X-rays (CXR) remain the first-line screening tool due to low cost and radiation, they suffer from limited sensitivity (roughly 70% visibility of lung volume due to anatomical superposition, in which all anatomical structures located in the path of the X-rays are stacked). [1]

CT is a radiographic technology that combines X-ray equipment with a computer and a cathode ray tube display to produce images of cross sections of the human body. A Radiographic detector measures the X-ray profile. Inside the CT scanner, there is a rotating frame that has an X-ray tube mounted on one side and the detector mounted on the opposite side. A beam of X-ray is generated as a rotating frame spins the X-ray tube and detector around the patient as shown in figure (4). Each time the X-ray tube and detector make one complete rotation, an image or slice is acquired. As the X-ray tube and detector make this rotation, the detector takes numerous profiles of the attenuated X-ray beam. Each profile is reconstructed by the computer into a 2D image of the slice that was scanned. [2]

What are Hounsfield Units (HU) and how are they used for tissue characterization? Key characteristics in CT -¿ GGO -¿ C/G ratio How do radiologists currently analyze these features manually?

2D -¿ 3D What is the typical structure and dimensions of a 3D chest CT volume? What are voxels and how do they differ from pixels? What preprocessing steps are typically required for CT data?

1.2 Problem Statement

Clearly define the problem the project aims to solve.

1.3 Objectives and Scope

State the objectives of your project clearly.

[1] Radiological Diagnosis in Lung Disease [2] A Comparative Study of Medical Imaging Techniques [3] A Review of Artificial Intelligence Integration in Medical Imaging

Chapter 2

Literature Review

2.1 Previous Work

Discuss relevant previous work in the field. For instance, [?] discusses advancements in Machine Learning.

2.2 Gaps in Literature

Identify gaps in the existing literature that your project aims to address.

Chapter 3

Methodology

3.1 Research Design

Describe the research design and methodology. Use diagrams or flow charts to illustrate the methodology.

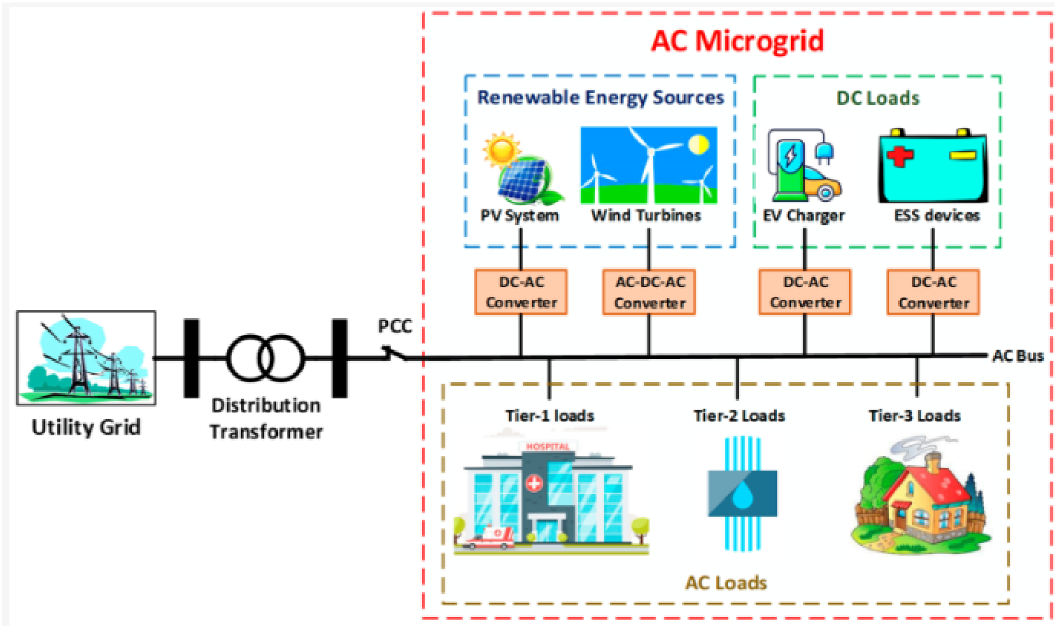

Figure 3.1: This is an example image.

As shown in Figure **??**, the image is centered and has a caption.

3.2 Data Collection

Explain how data will be collected for the project.

Chapter 4

Timeline and Resource Required

4.1 Timeline

Provide a realistic timeline for completing the project. Use a Gantt chart or table.

4.2 Resource Required

State the resource required for the project and estimate the budget for the resources required.

Chapter 5

Conclusion

Summarize the key points of your proposal and reiterate the importance of the project.